Big Issues in Economics and Finance

Coursework: Group Assignment 2026

Commodity price shocks, Growth and Inequality

This group assignment aims to give students the opportunity to use the concepts and methods explored in the lectures, in order to compare inequality and growth in two countries and examine how international energy and commodity price shocks affect their economies. You will combine quantitative analysis in R with economic reasoning to assess who is most vulnerable and evaluate a policy response.

Each group has been randomly allocated a pair of countries. The task of each group is to conduct the analysis described below for the two countries assigned and to present the findings of this analysis on 26 and 27 November 2026. The presentation slides and *R* code (with markup) used for the analysis are to be submitted in advance of the presentation (by 9:00 am 23 November 2026 via Canvas).

A. Objectives

Use data from the World Inequality Database (WID), alongside the international commodity-price and national economic data sets described in Section C, to complete the following tasks. The assessment is marked out of 100:

1. Visualize, describe and compare the evolution of income inequality and wealth inequality in the two countries (for the years for which data are available) using the programming language *R*. **[10 marks]**
2. Compare and contrast inequality for each country relative to the wider region (South Asia, Central America, MENA, etc) **[10 marks]**
3. Use *R* to assess the correlation between inequality (in income and wealth) and income growth. Groups should discuss reasons why ascertaining a causal nexus between these variables is difficult. **[20 marks]**
4. Use R to analyse the relationship between changes in international energy prices and domestic inflation in the two countries. Compare the timing and magnitude of the movements. Use food or energy consumer-price measures where available, and include recent developments in 2026 alongside a longer period. **[15 marks]**
5. Explain why energy and food price shocks may affect the two countries differently, considering their trade exposure, exchange rates and domestic policies. Assess which households and economic sectors are likely to be most vulnerable, drawing on your inequality findings and additional evidence. **[20 marks]**
6. Recommend one policy for each country to protect the vulnerable households from commodity-price shocks. Evaluate its distributional effects, fiscal cost and implications for inflation and incentives. **[15 marks]**
7. Briefly discuss the limitations of the data and analysis. **[10 marks]**

B. Deliverables and presentation

A. Presentation materials

Slides and R code to be submitted by **09:00 am 23 November 2026**. The number of slides included in the presentation should not exceed 16 (including the title slide) and the number of graphs should not exceed 10. Only choose figures that directly support the comparison and your conclusions.

B. Presentation

Each group will present for 10 minutes, followed by approximately 7 minutes of discussion, with questions from the *module leader*, the *tutorial leader* and assigned discussants (further details below).

> Presentations should be delivered by 2 or 3 members of each group (although all can present if you so choose) – it is up to each group to choose which members will deliver the presentation. The remaining members of the group, who do not deliver the presentation, will need to actively partake in answering questions during the Questions and answer section. We reserve the right to amend an individual's mark if they are deemed to have not participated effectively during the presentation. On the day of the presentation, students arriving late or leaving during the presentation may incur an individual/group penalty. **Presenters must deliver their presentation without cue cards, written notes or electronic devices used as prompts. This includes phones, tablets, laptops and smartwatches used to read notes or a script.**
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> Prepare to explain the findings in your own words. The presentation should communicate the economic argument rather than read out the text on the slides.

Presentations take place on **26 and 27 November 2026**.

C. Discussion

Each group presentation is followed by approximately 7 minutes of discussion, with questions from the module leader, the tutorial leader, and members of another group who will serve as *discussants*. Discussants are randomly allocated the discussion of the presentation delivered by another group. They will be provided with that group's presentation slides shortly after these are submitted to the Programmes Team.

Discussants should review the slides and analysis of the group they have been assigned and come up with one or, time-permitting, at most two questions to pose to the group presenters on the day of the presentation. The pertinence of such questions will be assessed and will contribute towards the overall mark attained by the discussants' group as an adjustment to their presentation mark (either positive or negative, capped at + or - 10).

C. Data and analytical guidance

Inequality and growth

Use WID (https://wid.world/) for the core inequality analysis. Select comparable income and wealth concepts, population definitions and percentile groups for both countries and their regional benchmarks. The top 10% and bottom 50% shares can provide a clear starting point. Report the definitions used and any gaps in coverage.

Use a consistent real income-per-person measure to calculate income growth. If using GDP per capita from another source, identify it clearly and explain how it differs from the WID income concept. Refer to the R examples provided in the course for support.

Commodity prices and domestic inflation

World Bank Commodity Markets and Pink Sheet: https://www.worldbank.org/en/research/commodity-markets. Download historical monthly energy prices, such as Brent crude oil, and relevant food prices or indices. Use observed prices for the empirical analysis; identify forecasts separately.Where a sufficiently long monthly series is unavailable, quarterly analysis is acceptable (but you may wish to extend the analysis further back).

IMF Consumer Price Index dataset: https://data.imf.org/Datasets/CPI. Obtain national headline CPI and, where available, food or energy-related components. Official national statistical offices may be used where international data are incomplete. Document any differences in coverage or definitions.

Country exposure and supporting evidence

World Development Indicators: https://databank.worldbank.org/source/world-development-indicators. Relevant measures include fuel exports as a share of merchandise exports, fuel imports as a share of merchandise imports and GDP per capita. Check the observation year before using a measure of exposure.

IMF Commodity Terms of Trade: https://data.imf.org/en/datasets/CTOT. These country-specific measures can help explain exposure to changing international commodity prices. They are optional supporting data, rather than a requirement.

Preparing and comparing the data

- For the topical analysis, aim to use monthly data from 2010 to the latest common available month in 2026. If coverage is shorter, use a defensible common period and explain the choice.
- For Objective 4, compare percentage changes in commodity prices with inflation rates. State whether changes are monthly or year-on-year and keep definitions consistent. If showing price levels, rebase indices to a common starting value.
- Align observations by date. Do not repeat annual inequality observations across months to create an artificial monthly sample. Identify missing values, estimates and revisions, and do not treat unavailable observations as zeros.

Once the dataset is constructed, one can start conducting the analysis of objectives 1 to 7, described above (Section A). The data visualisations using *R* we saw in Lecture 2 can be useful at this stage (despite the countries and the purpose of the analysis being different).

D. Support in the completion of the coursework assignment

Over the period leading to the completion of the assignment, support is available during the module leader and tutorial leader's office hours to discuss the economics / finance involved in the assignment.

E. Marking and feedback

Marks and written feedback will be provided to each group shortly after the presentation. Please note that this coursework assignment is groupwork, so each member of the group is expected to receive the same mark. However, individual students' marks may be individually adjusted if necessary (for example, if they do not contribute to the group project adequately or arrive late for the presentation slot).

The marking scheme is described in Section A. For each of the 7 sub-tasks we will assess:

- Quality of the analysis. Are the facts (trends, correlations, etc) correct and is the interpretation of these facts adequate? Are the hypotheses raised logically coherent? Are the policy suggestions sensible?
- Clarity of the presentation: are the slides clear? Are they presented clearly? Are all the members of the group able to answer questions effectively? Are the questions asked to other groups sensible?
- Group cooperation: did the group work collaboratively and cohesively to complete the task?

All students are expected to be present for all presentations in their allotted tutorial group. Students arriving late for the allotted tutorial time slot may incur a mark penalty depending on the severity of the delay (capped at -5 marks out of 100 for the individual/group).

F.FAQs

- The use of generative AI (for example, ChatGPT) is permitted, encouraged and can be helpful in the coding aspect of the task. However, this needs to be fully declared: if you use an AI tool, please add a Word document to your submission indicating all your AI prompts and explaining how you used them (what, for what purpose, results obtained). Undeclared use of generative AI is tantamount to academic misconduct. Declared use is fine. You remain responsible for the accuracy of the data, code, references and conclusions. All group members must be able to explain the analysis and answer questions about it.

- References: it is important to cite all academic work referenced in your slides. Any standard referencing style is fine, provided that this is used consistently throughout the

assignment. References should be listed in the last slide of the presentation (and will not count towards the slide limit)

- It is part of the assessment to work collaboratively within your groups. If any group has concerns about *free riding* they should reach out to the module/tutorial leader as soon as possible. Raising concerns close to the presentation date will not give us sufficient time to intervene.